



MMWAVE RadarSS Release Notes

1 Introduction

RadarSS firmware is responsible for configuring RF/analog and digital front-end in real-time. It also schedules periodic temperature based calibrations and monitors based on use need. This enables the mm-Wave front-end to be autonomous and capable of adapting itself to handle temperature and ageing effects, and to enable significant ease-of-use.

2 Release Overview

2.1 Platform and Device Support

The device and platforms supported with this release include:

Supported Devices	Release Status	Supported EVMs
xWR254x ES1.0	Production Release	xWR254x EVM

2.2 Release version

Version	Type
ROM: 2.5.4.0 PATCH: 2.5.8.1	Binary patch (4.44KB)

3 Release Contents

3.1 xWR254x ES1.0 Patch contents

xWR254x ES1.0 has RadarSS Firmware contents in ROM and the below DFP releases provide patch on top of ROM. The below DFP release are applicable only for ES1.0 through firmware patch updates.

3.1.1 DFP 2.4.11.0 Features and Enhancements

- xWR254x is TI's 77GHz RF CMOS Radar SOC. Features supported in this firmware release:
 - xWR254x RadarSS ROM Firmware is derived from AWR2944 DFP2.4.9 release baseline with additional enhancements. It is backward compatible to DFP 2.4.9 supported APIs with minor updates for enhancement.
 - Supports 4 TX and 4 Real-only RX chains.
 - Synthesizer RF frequency supported 76 – 81GHz
 - VCO1: 76 – 77GHz
 - VCO2: 76 – 81GHz
 - Supports 7 functional profiles.
 - Supports 266MHz/us max slope
 - Supports programmable filter
 - Support for inter-chirp jitter mitigation
 - Synth frequency monitor (Live Mode) support for multiple profiles
 - Support for enabling 1 ns resolution for ADC start time in advanced chirp configuration
 - Added new FRC DCC clock monitor
 - Added support to disable parameters in advanced chirp configuration
 - Added support to enable/disable OSCCLKOUTETH to provide reference clock for external ethernet PHY
- Refer mmWave-Radar-Interface-Control.pdf (ICD) for more information on xWR254x ES1.0 API details.

3.1.2 DFP 2.4.12.0 Features and Enhancements

- Improvements in the LODIST calibration
Fix for monitoring time stamps when RSS dynamic frequency switch is enabled and working with 50MHz XTAL

3.1.3 DFP 2.4.12.1 Features and Enhancements

- No changes from DFP 2.4.12.0

3.1.4 DFP 2.4.13.0 Features and Enhancements

- Adjusted synth control voltage monitor and fault injection for robustness
- Enabled debug support for LPF monitor in RX IFA Stage monitor
- Improved operation for inter-chirp jitter mitigation mode
- Improved temperature sensor accuracy

3.1.5 DFP 2.4.14.0 Features and Enhancements

- Minor bug fixes

3.1.6 DFP 2.4.17.0 Features and Enhancements

- No changes to patch binary from DFP 2.4.14.0
- Refer 3.2.6 in Feature/Change list and 4.1.1 in Unsupported features and APIs for documentation updates

3.2 Feature/Changes List

3.2.1 Changes in DFP 2.4.11.0 release (with respect to DFP 2.4.9.1 / AWR294x)

Item type	Key	Description
Feature	MMWAVE_DFP-1888	Synth frequency monitor (Live Mode) support for multiple profiles
Feature	MMWAVE_DFP-1886	Support for enabling 1 ns resolution for ADC start time in advanced chirp configuration
Feature	MMWAVE_DFP-1884	Added new FRC DCC clock monitor
Feature	MMWAVE_DFP-1888	Added support to reduce inter-chirp jitter

Feature	MMWAVE_DFP-1882	Added support to disable parameters in advanced chirp configuration
Feature	MMWAVE_DFP-1880	Added support to enable/disable OSCCLKOUTETH to provide reference clock for external ethernet PHY

3.2.2 Changes in DFP 2.4.12.0 release (with respect to DFP 2.4.11.0)

Item type	Key	Description
Bug	MMWAVE_DFP-2005	Fix for LODIST calibration failure at 80 deg C
Bug	MMWAVE_DFP-2007	Fix for monitoring time stamps when RSS dynamic frequency switch is enabled with 50MHz XTAL

3.2.3 Changes in DFP 2.4.12.1 release (with respect to DFP 2.4.12.0)

No change in RadarSS firmware

3.2.4 Changes in DFP 2.4.13.0 release (with respect to DFP 2.4.12.1)

Item type	Key	Description
Bug	MMWAVE_DFP-2015	Fix for synth control voltage monitor failure when fault injection is enabled
Bug	MMWAVE_DFP-2018	Enable debug support for LPF monitor in RX IF Stage monitor
Bug	MMWAVE_DFP-2021	Fix for synchronization in inter-chirp mitigation mode
Bug	MMWAVE_DFP-2025	Fix to generate error when OSCCLKOUT and OSCCLKOUTETH are enabled at the same time
Bug	MMWAVE_DFP-2022	Improved temperature sensor accuracy
Bug	MMWAVE_DFP-2023	Timing updates for chirp and monitoring

3.2.5 Changes in DFP 2.4.14.0 release (with respect to DFP 2.4.13.0)

Item type	Key	Description
Bug	MMWAVE_DFP-2065	Fix for synthesizer frequency monitor failure at cold temperatures.
Bug	MMWAVE_DFP-2051	Fix for SYNTH_VCO_OPENLOOP fault injection failure at cold temperatures.
Bug	MMWAVE_DFP-2069	Fix for Synth VCO LDO short circuit fault at hot temperatures
Bug	MMWAVE_DFP-2085	Fix for RX Gain accuracy improvement

3.2.6 Changes in DFP 2.4.17.0 release (with respect to DFP 2.4.14.0)

Item type	Key	Description
Bug	MMWAVE_DFP-2144	OSCLKOUT is not supported in xWR254x

4 Unsupported features and APIs (applicable to all DFP releases)

The following APIs and features are not validated fully at system level, it is recommended not to use these APIs in this and all previous DFP releases. This list of unsupported features is in addition to the list mentioned in known issues.

4.1 Unsupported Features

4.1.1 Functional APIs

API	Feature	Description
PCR self-test feature in AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB	PCR self-test	The PCR self-test is not supported in xWR254x.
AWR_INTERCHIRP_BLOCKCONTROLS_SB	Inter-chirp power saving timing configurations	This API is not validated at system level. It is recommended not to use the same.
AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB	RX noise figure monitor	The RX noise figure monitor is not supported in xWR254x.
AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB	Rx mixer input power monitor	The RX mixer input power monitor is not supported in xWR254x.
TXn POWER_BACKOFF fields in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB	Calibration and Monitoring Frequency and TX Power limits API	Recommended to perform factory calibration with 0dB back-off set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API and, store and restore the calibration data from non-volatile memory. With this option, user can set nonzero back-off option in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API during normal runs to back-off monitor/calibration chirp's power.
AWR_MONITOR_VMON_CONF_SB	VMON configuration	This feature has not been validated at a system level. It is not recommended to use this.
SYNTH_VCO1_SLOPE and SYNTH_VCO2_SLOPE fields in the PLL_CONTROL_VOLTAGE_VALUES in AWR_MONITOR_PLL_CONTROL_V	VCOx slope monitoring	This feature is not supported in xWR254x.

OLTAGE_REPORT_AE_SB		
LPF_CUTOFF_BANDEDGE_DROOP_VALUE_RX0, LPF_CUTOFF_STOPBAND_ATTEN_VALUE, LPF_CUTOFF_BANDEDGE_DROOP_VALUE_RX fields in AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB ASYNC report	LPF monitoring results ASYNC report	These fields are only supported for debug purposes and should not be used for production use-cases.
LPF_CUTOFF_BANDEDGE_DROOP_THRESH and LPF_CUTOFF_STOPBAND_ATTEN_THRESH fields in AWR_MONITOR_RX_IFSTAGE_CONFIG_SB API	LPF monitoring threshold fields in the IF STAGE monitoring configuration API	These fields are only supported for debug purposes and should not be used for production use-cases.
Some combinations of RX_GAIN and RF_GAIN_TARGET in PF_RX_GAIN field of the AWR_PROFILE_CONF_SB and AWR_CONT_STREAMING_MODE_CONFIG_SET_SB APIs.	(RX Gain, RF Gain Target) (44dB, 32dB)	Since the maximum IF gain is +10dB in xWR294x, the specified pairs of (RX Gain, RF Gain Target) settings are not supported.
Some values of the GLOBAL_RESET_MODE field of the AWR_ADVANCE_CHIRP_CONFIG_SB API are not supported when using Advanced Frame Configuration.	Value 0 - "End of frame"	When using advanced frame configuration in conjunction with advanced chirp configuration, the "end of frame" reset mode is not supported in advanced chirp config API.
b27 - Bus Safety test in field RF_POWERUP_BIST_STATUS_FLAGS of AWR_RF_BOOTUPBIST_STATUS_GET_SB API	Bus Safety test Status	This feature is not supported in xWR254x.
b30 - BUS_SAFETY_PCR_ERROR b31 - BUS_SAFETY_BSS_SLV_ERROR in field ESM_GROUP1_ERRORS_LSB and b0 - BUS_SAFETY_BSS_MST_ERROR b1 - BUS_SAFETY_STATIC_MEM_ERROR b2 - BUS_SAFETY_MBOX_ERROR b3 - BUS_SAFETY_SEC_AGGREGATED_ERROR in field ESM_GROUP1_ERRORS_MSB of AWR_RF_ADV_ESMFAULT_STATUS	Bus Safety ESM Error	This feature is not supported in xWR254x.

S_GET_SB API b30 - BUS_SAFETY_PCR_ERROR b31 - BUS_SAFETY_BSS_SLV_ERROR in field ESM_GROUP1_ERRORS_LSB and b0 - BUS_SAFETY_BSS_MST_ERROR b1 - BUS_SAFETY_STATIC_MEM_ERRO R b2 - BUS_SAFETY_MBOX_ERROR b3 - BUS_SAFETY_SEC_AGGREGATED_ ERROR in field ESM_GROUP1_ERRORS_MSB of AWR_AE_RF_ADV_ESMFAULT_SB API		
b24 – FFT test in field RF_POWERUP_BIST_STATUS_FL AGS of AWR_RF_BOOTUPBIST_STATUS_G ET_SB API b24 – FFT test in field DIG_MONITORING_ENABLES of AWR_MONITOR_RF_DIG_LATENTF AULT_CONF_SB API b24 – FFT test in field DIG_MON_LATENT_FAULT_STATUS of AWR_MONITOR_RF_DIG_LATENTF AULT_REPORT_AE_SB API	FFT test	This feature is not supported in xWR254x.
b4 - DFE_PARITY_AND_OUT_ERROR in field ESM_GROUP1_ERRORS_LSB of AWR_RF_ADV_ESMFAULT_STATU S_GET_SB API	DFE AND Parity ESM Error	This feature is not supported in xWR254x.

b4 - DFE_PARITY_AND_OUT_ERROR in field ESM_GROUP1_ERRORS_LSB of AWR_AE_RF_ADV_ESMFAULT_SB API		
b0 - Digital RX gain compensation enable in field DIGITAL_COMP_EN and DIGITAL_RX_GAIN_COMP field of AWR_DIGITAL_COMP_EST_CONTR OL_SB API	Digital RX gain compensation	This feature is not supported in xWR254x.
OSCCLKOUT_DIS in CASCADING_ PINOUTCFG field of the AWR_CHAN_CONF_SET_SB API	OSCCLKOUT	OSCCLKOUT is not supported. It is recommended to keep OSCCLKOUT disabled.

4.1.2 Debug APIs

API	Feature	Description
AWR_RF_PALOOPBACK_CFG_SB AWR_RF_LOLOOPBACK_CFG_SB AWR_RF_IFLOOPBACK_CFG_SB	Loopback enables	PA, LO and IF loopback APIs are not supported in functional mode, recommended to use only for debug.
AWR_RF_TEST_SOURCE_CONFIG_ SET_SB AWR_RF_TEST_SOURCE_ENABLE_ SET_SB	Test source feature	Test source feature is not supported in functional mode, recommended to use only for debug.
AWR_CONT_STREAMING_MODE_ CONF_SET_SB AWR_CONT_STREAMING_MODE_ EN_SB	Continuous streaming mode	Continuous streaming mode is not supported in functional mode, recommended to use only for debug.

5 Known Issues

Key	Severity	Description
MMWAVE_DFP-1649	Minor	Noise metric indicators vary with temperature. RF loopback-based TX and RX gain phase mismatch monitors include a noise metric indicator to enable detection of corruption by external interference. The noise metric varies with temperature. Strategy to determine thresholds for detecting interference is mentioned in a monitoring application note. The strategy includes temperature dependent thresholds.

MMWAVE_DFP-2105	Major	ECC_AGC_DED_ERROR mapped to ESM_GROUP1_ERRORS_LSB Double bit ECC errors in BSS static RAM and BSS mailbox indicated by ECC_AGC_DED_ERROR is mapped to ESM_GROUP1_ERRORS_LSB. ESM_GROUP1_ERRORS are group of non-fatal errors, however double bit ECC error ECC_AGC_DED_ERROR is a fatal error. Workaround: It is recommended to treat ECC_AGC_DED_ERROR in field ESM_GROUP1_ERRORS_LSB of AWR_AE_RF_ADV_ESMFAULT_STATUS_SB and AWR_RF_ADV_ESMFAULT_STATUS_SB should be treated as a fatal error
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